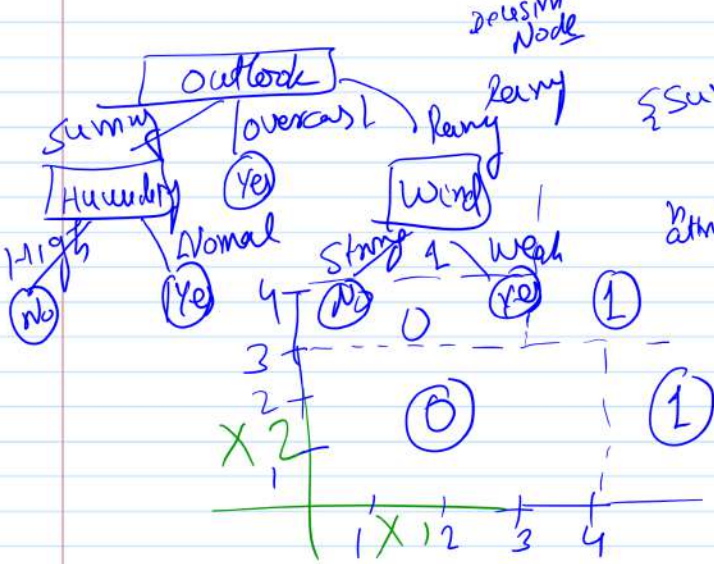


COL774
OCT 7, 2025

Decision Trees - (Internal)



$$\text{Data} = \{x^u, y^u\}_{u=1}^m$$

Grow Tree ($\mathcal{D}$)

if $\forall u \in \mathcal{D}, y^u = k$, return Leaf(k);

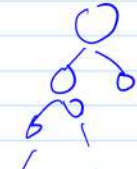
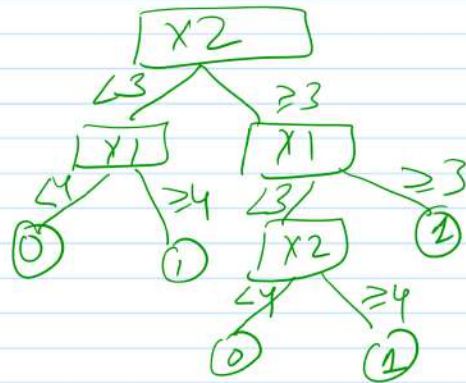
$X_d = \text{ChooseAttrToSplit}(\mathcal{D})$;

$\mathcal{D}_e = \{x^u, y^u\} : x_d^u = e\}$

Return ($X_d, \{\text{GrowTree}(\mathcal{D}_e)\}_{e=1}^{|V|}$)

It is organized in the form of a decision tree $\{H, N, C\}$

Attribute	Sunny	Normal	Sunny	Normal	Play (Y/N)
Outlook					
Humidity					
Wind					
Temp					
Play (Y/N)					



$X_d \in \mathbb{I}$
 $e \in \{1 \dots |V|\}$

